

INFO6007 Practice Question Bank: Weeks 1–7

(With Answer Guide)

Instructions for Practice Questions (Preparation for Interactive Oral/Viva)

- The **interactive oral (viva)** will take place during **tutorial sessions in Week 8, 9 and 10**.
- It is worth **10%** of your final grade, will run for **7-10 minutes per student**, and will cover **topics from Weeks 1–7**.
- **AI tools are strictly prohibited** during the viva.

We have provided a **Practice Question Bank** with an **Answer Guide**.

How to use these practice questions and guides:

1. Review the lecture and tutorial materials for each week before attempting the questions.
2. Use the **Practice Bank** as a warm-up to check your understanding and confidence.
3. The **Answer Guide is not a script to memorise**, it gives you the key points expected in a strong answer.
 - Add your own examples (e.g., case studies, workplace scenarios, news items).
 - Rephrase the ideas in your own words so you sound natural.
4. Practice answering questions **aloud** within 1–2 minutes each. This mirrors the viva style.
5. Work with a partner or group: take turns asking questions and providing answers, then compare with the guide.
6. Focus on **clarity, logical flow, and application** of concepts, not length. Quality and depth matter more than quantity.

Practice Question Bank – Week 1

1. **Define a project and explain its key characteristics.**

Answer Guide: A project is a temporary endeavor to create a unique product, service, or result. Its key characteristics include being temporary (with start and end points), producing a unique outcome, having specific objectives, and operating within constraints such as scope, time, cost, and resources.

2. **Why are IT projects considered more complex than other projects?**

Answer Guide: IT projects involve rapidly evolving technologies, require specialized

skills, deal with interdependent systems, and must align closely with organizational goals.

3. **List three consequences of not applying IT project management.**

Answer Guide: Without project management, teams risk scope creep, missed deadlines, budget overruns, technical failures, and stakeholder dissatisfaction.

4. **Differentiate between project, program, and portfolio management.**

Answer Guide: A project is a single effort, a program is a collection of related projects managed together for benefits, and a portfolio is a higher-level grouping of projects and programs aligned with strategic objectives.

5. **Mention two core skills and two soft skills of an IT project manager.**

Answer Guide: Core skills: **planning & scheduling, cost control**. Soft skills: communication, leadership, negotiation, adaptability.

6. **What role does **stakeholder management** play in IT project success?**

Answer Guide: It ensures expectations are captured, reduces resistance, builds buy-in, and improves project alignment with business goals.

7. **What are “scope, time, cost, quality” known as in PM?**

Answer Guide: These form the “triple (or quadruple) constraints” — interdependent project dimensions.

8. **Why is communication considered critical in IT project management?**

Answer Guide: Clear communication fosters coordination, supports stakeholder engagement, reduces misunderstanding, and enables effective decision-making.

Practice Question Bank – Week 2

1. **What is a project lifecycle? Name its phases.**

Answer Guide: The project lifecycle is a structured sequence of phases: Initiation, Planning, Execution, Monitoring & Controlling, and Closure.

2. **Differentiate between Waterfall and Agile methodologies.**

Answer Guide: Waterfall is linear and sequential, with strict phases, while Agile is iterative and adaptive, focusing on incremental delivery and frequent feedback.

3. **Provide one real-world example where Waterfall methodology is preferable.**

Answer Guide: Government payroll systems or compliance-heavy projects where requirements are fixed and extensive documentation is required.

4. **What are two strengths of Agile in IT projects?**

Answer Guide: Ability to adapt to changing requirements and incorporating regular customer feedback.

5. **Describe Scrum roles.**

Answer Guide: Product Owner manages the backlog, Scrum Master facilitates processes and removes blockers, and Scrum Team delivers the work in sprints.

6. **What is the purpose of a Kanban board?**

Answer Guide: To visualize work stages, track tasks, manage WIP limits, and ensure smooth workflow.

7. **How does DevOps complement Agile?**

Answer Guide: By integrating development and operations, automating CI/CD pipelines, improving collaboration, and ensuring continuous delivery with speed and security.

8. **Name two techniques used for requirements gathering.**

Answer Guide: Interviews and workshops; others include surveys, prototyping, and observation.

Practice Question Bank – Week 3

1. **What is a Scope Management Plan?**

Answer Guide: A document that defines how scope will be planned, validated, and controlled throughout the project.

2. **Define Work Breakdown Structure (WBS).**

Answer Guide: A hierarchical decomposition of project deliverables into manageable work packages that guide scheduling and resource allocation.

3. **Why is scope validation important?**

Answer Guide: It provides formal acceptance of deliverables, prevents disputes, and minimizes scope creep.

4. **Mention three common issues in scope management.**

Answer Guide: Scope creep, lack of stakeholder engagement, and miscommunication about what is included/excluded.

5. **What is schedule management?**

Answer Guide: Planning, developing, managing, and controlling the project timeline to ensure on-time completion.

6. **List two techniques for estimating activity duration.**

Answer Guide: Analogous estimating and three-point estimating (PERT); others include parametric and expert judgment.

7. **What is a schedule baseline?**

Answer Guide: The approved schedule version **used for tracking project progress** against planned timelines.

8. **What happens when overambitious scope is defined?**

Answer Guide: It increases complexity and risk, leading to delays, cost overruns, and higher likelihood of project failure.

Practice Question Bank – Week 4

1. **Define project cost management.**

Answer Guide: The process of estimating, budgeting, and controlling costs to ensure project delivery within approved financial resources.

2. **Differentiate between direct and indirect costs.**

Answer Guide: Direct costs are directly tied to deliverables (e.g., developer salaries). Indirect costs support overall operations (e.g., office rent, utilities).

3. **What is the purpose of reserves in cost management?**

Answer Guide: To account for uncertainties. Contingency reserves cover identified risks, while management reserves cover unknown risks.

4. **Describe bottom-up estimating.**

Answer Guide: Estimating costs at the task/work package level, then aggregating for the total — **most accurate but time-consuming.**

5. **Provide an example of three-point estimation.**

Answer Guide: Cloud infrastructure setup: Optimistic = \$8K, Likely = \$12K, Pessimistic = \$20K → $(O + 4M + P) / 6 = \$12.6K$.

6. **What is Earned Value Management (EVM)?**

Answer Guide: A performance measurement methodology integrating scope, schedule, and cost to track project health.

7. **Explain the difference between Planned Value (PV) and Earned Value (EV).**

Answer Guide: PV is the budgeted cost for planned work, while EV is the budgeted cost of actual work completed.

8. **What are common mistakes in IT cost estimation?**

Answer Guide: Ignoring integration/training costs, overestimating efficiency, not accounting for maintenance, and vendor lock-in.

Practice Question Bank – Week 5

1. **Define quality in IT projects.**

Answer Guide: The degree to which deliverables meet requirements, expectations, and are fit for intended use.

2. **Differentiate QA vs QC.**

Answer Guide: QA ensures processes prevent defects (proactive). QC verifies deliverables meet standards (reactive).

3. **What are the 4 principles of quality management?**

Answer Guide: Customer focus, prevention over inspection, continuous improvement, and fact-based decision making.

4. **Name two IT quality standards.**

Answer Guide: ISO 9001 for quality systems; ISO/IEC 27001 for information security.

5. **What does CMMI stand for?**

Answer Guide: Capability Maturity Model Integration; framework for process improvement in software development.

6. **Explain Verification vs Validation.**

Answer Guide: Verification = did we build it right? Validation = did we build the right thing?

7. **Mention two Agile perspectives on quality.**

Answer Guide: Shared responsibility for quality and continuous testing with frequent feedback.

8. **Give one example of risk and defect prevention in IT projects.**

Answer Guide: Using static analysis tools like SonarQube to detect vulnerabilities before release.

Practice Question Bank – Week 6

1. **Define project resource management.**

Answer Guide: Identifying, acquiring, and managing human and physical resources for successful project delivery.

2. **Differentiate leadership vs management.**

Answer Guide: Leadership inspires and motivates; management organizes, plans, and controls.

3. **What is intrinsic vs extrinsic motivation?**

Answer Guide: Intrinsic = self-driven, curiosity/interest; Extrinsic = external incentives such as money or promotions.

4. **List Maslow's hierarchy of needs in PM context.**

Answer Guide: Physiological → Safety → Belonging → Esteem → Self-actualization; managers must meet lower needs first.

5. **What is a RACI matrix?**

Answer Guide: A responsibility assignment chart: Responsible, Accountable, Consulted, Informed.

6. **What is Resource Breakdown Structure (RBS)?**

Answer Guide: A hierarchical decomposition of resources (human, equipment, tools, materials).

7. What does a resource histogram show?

Answer Guide: A bar chart showing resource allocation and workload distribution over time.

8. Name three techniques for estimating activity resources.

Answer Guide: Expert judgment, analogous estimating, bottom-up estimating, and three-point estimating.

Practice Question Bank – Week 7

1. Explain the purpose of Project Integration Management in IT projects.

Answer Guide: Integration ensures all processes and deliverables align. In IT, it means coordinating code modules, third-party tools, databases, and ensuring everything works together toward the business goal.

2. Imagine your IT project is facing scope creep. How would you manage it?

Answer Guide: Apply change control, assess cost/time/resource impacts, communicate with stakeholders, and only approve if aligned with business goals.

3. Why do we use a Work Breakdown Structure (WBS) in IT project scope management?

Answer Guide: It breaks the project into smaller tasks (e.g., frontend, backend, database setup) so responsibilities are clear and progress is measurable.

4. Describe a real example of a risk in an IT project and how you would handle it.

Answer Guide: Example: Vendor server downtime → Mitigate by having backup hosting or cloud redundancy.

5. What role does Quality Assurance play in IT project success?

Answer Guide: Ensures deliverables meet requirements and function correctly through testing, reviews, and continuous integration pipelines.

6. How do project schedule management techniques (like Critical Path Method) help IT project managers?

Answer Guide: They highlight tasks that cannot be delayed without delaying the project—e.g., coding must finish before system testing.

7. Why is stakeholder management more complex in IT projects than in other industries?

Answer Guide: Because multiple groups (developers, clients, users, vendors, executives) have different technical understanding, goals, and expectations.

8. What is the importance of communications management in IT project success?

Answer Guide: Prevents misunderstandings, aligns technical and business teams, ensures timely updates, and builds trust.

Marking Guide

The marking criteria below are shown out of 100 marks.

The final mark will be scaled down to 10 marks.

Criteria	Marks
Knowledge & Understanding of Project Management Concepts	30
Application to IT Project Scenario	20
Communication, Structure & Clarity	20
Critical Thinking & Problem Solving	15
Professionalism & Engagement (including individual contribution)	15
Total	100